

SIMATS ENGINEERING
C02-ASSESSMENT 1
ITA1407-ETHICAL HACKING
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COURSE OUTCOME COVERED

CO2: Foot printing Tools: Conducting Competitive Intelligence, DNS Zone Transfers, Introduction to

Social Engineering: Social Engineering Attacks and Countermeasures. (BT3)

Report Analysis

Questions

1. A security analyst performed a network scan on the target 192.168.1.10 using the Nmap tool. The report identified several open ports, including HTTP (80), SSH (22), and MySQL (3306), along with the operating system and service versions. Analyse the report and explain the security risks associated with the identified services. Discuss how an attacker could exploit these services and recommend suitable security measures to protect the target system.

ANSWER

Nmap Network Scan (20 Marks)

Command

nmap -sV -O 192.168.1.10

Explanation

- nmap → Starts the Nmap scanner.
- -sV → Detects service versions.
- -O → Detects the operating system.
- 192.168.1.10 → Target IP address.

Example Output

```
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.2
80/tcp    open  http     Apache httpd 2.4.41
3306/tcp  open  mysql    MySQL 8.0.28
OS Details;Linux Ubuntu 20.04
```

2. A phishing awareness campaign was conducted for employees of ABC Technologies Pvt. Ltd. Using the GoPhish tool. The report indicates that 40% of employees clicked the phishing link, 20% entered their login credentials, and several users downloaded a malicious attachment. Analyze the report and identify the security weaknesses demonstrated by the employees. Recommend suitable user awareness programs, email security controls, and organizational policies to reduce the risk of successful social engineering attacks.

ANSWER

GoPhish Phishing Campaign (20 Marks)

Start GoPhish

./gophish

Access Dashboard

<https://127.0.0.1:3333>

Configure

- Create Sending Profile
- Create Landing Page
- Create Email Template
- Import Employee List
- Launch Campaign

Example Report

Emails Sent: 100

Emails Opened: 70

Clicked Link: 40%

Credentials Submitted: 20%

Attachments Downloaded: 10

3. A DNS assessment report generated using the Dig tool shows that the organization's DNS server permits a successful DNS Zone Transfer (AXFR), exposing internal hostnames, mail servers, and DNS records. Analyse the report and explain the security implications of the exposed information. Discuss how attackers could exploit these findings and recommend best practices to secure the DNS infrastructure

ANSWER

Command

```
dig AXFR example.com @ns1.example.com
```

Alternative

```
dig example.com AXFR
```

Example Output

```
example.com.
```

```
mail.example.com.
```

```
ftp.example.com.
```

```
vpn.example.com.
```

```
server1.example.com.
```

```
www.example.com.
```